

Galaxy Pipelines for Bacteria & Virus Analysis

Catalog + a first pass at description-to-pipeline mapping

Talk to Galaxy · NIAID-BRCs AI Codeathon 2.0

Source: IWC registry — github.com/galaxyproject/iwc

Why this matters

Goal 1 of the proposal: **route before building**.

Determine, early in the conversation, whether a published workflow already does what the researcher needs — **use**, **adapt**, or **build**.

This deck is that decision surface, scoped to bacteria & virus workflows:

- What actually exists in the IWC registry today
- Machine-readable catalog: `knowledge_base/pathogen_genomics/galaxy_pipeline_catalog.yaml`
- A working v0 matcher: `scripts/search_pipeline_catalog.py`

Bacterial Genomics (4 workflows)

Workflow	Input	Does
cgMLST	Assembled contigs	Strain typing via curated MLST schemes (pubMLST, Enterobase, cgMLST.org)
AMR Gene Detection	Assembled genome	StarAMR + AMRFinderPlus + ABRicate → resistance & virulence genes
Genome Annotation	Assembled genome	Bakta CDS calling, integrons, plasmids, insertion sequences
QC & Contamination Control	Illumina PE FASTQ	Quast/Checkm2 assembly QC, Kraken2/Bracken taxonomic ID

`workflows/bacterial_genomics/` in IWC

Virology (3 workflows)

Workflow	Input	Does
Generic Non-Segmented Viral Variant Calling	Illumina PE, $\pm$ amplicon	fastp→bwa-mem→lofreq/ivar→SnpEff for simple viruses (e.g. Morbillivirus)
Influenza A Subtyping + Consensus	Illumina PE	VAPOR segment matching, HA/NA subtyping, consensus, phylogenetics
Pox Virus Amplicon (half-genome)	Illumina PE, 2 runs/sample	Resolves ITRs by merging masked-reference mappings, iVar consensus

workflows/virology/ in IWC

SARS-CoV-2 (7 workflows)

Workflow	Data	Purpose
WGS PE / SE Variant Calling	Illumina WGS	bwa-mem/bowtie2 + lofreq + snpEff
Illumina Amplicon ARTIC (lofreq)	Illumina amplicon	Ampliconic variant calling, primer-site QC
Illumina Amplicon ARTIC (iVar)	Illumina amplicon	Rapid majority variants + pangolin/nextclade lineage
ONT ARTIC Variant Calling	Nanopore amplicon	minimap2 + medaka, ARTIC-style
Consensus Construction	VCF + BAM	Builds consensus from any of the above
Variation Reporting	VCF	Tabular + allele-frequency reports

`workflows/sars-cov-2-variant-calling/` in IWC — the most mature pipeline family

Metagenomics & Pathogen ID (6 workflows)

Workflow	Organism scope	Does
Nanopore Preprocessing	agnostic	QC + host-read removal
Taxonomy Profiling (Krona)	agnostic	Kraken2 species ID + visualization
Gene-based Pathogen ID	agnostic	Virulence factor + AMR gene screening
Allele-based Pathogen ID	agnostic	SNP-based strain/variant tracking
PathoGFAIR Aggregation	agnostic	Cross-sample heatmaps, phylogenies
Raw Read AMR (metagenomic)	bacteria-leaning	Sylph taxonomy + Groot/deepARG AMR from raw short reads

workflows/microbiome/ in IWC — useful when the organism is *unknown*

Coverage snapshot: 20 workflows, honestly thin in places

- **Deep coverage:** SARS-CoV-2 (7 dedicated workflows)
- **Moderate coverage:** generic bacterial genomics (typing, AMR, annotation, QC)
- **Sparse coverage:** named viruses beyond SARS-CoV-2 — only Influenza and Pox virus have dedicated workflows
- **No dedicated workflow for:** most WHO priority pathogens by name (e.g. no "Mtb workflow," no "Salmonella workflow" — bacterial workflows here are genus-agnostic)

Implication: many real requests will legitimately route to **adapt** (a close generic workflow) or **build**, not **use**. That's a finding for the pass-rate report, not a catalog gap to silently paper over.

Can we map a free-text description to these pipelines?

Yes — a first version exists and works.

```
scripts/search_pipeline_catalog.py + knowledge_base/pathogen_genomics/galaxy_pipeline_catalog.yaml
```

Approach (v0, intentionally simple and auditable):

1. Each catalog entry carries `organism`, `data_type`, `analysis_type`, and free-text `keywords`
2. Tokenize the researcher's description
3. Score each workflow by keyword-token overlap
4. Return ranked candidates **with the matched terms shown** — no black box

This is a keyword router, not semantic search — the honest v0 this catalog matcher still is, even now that a real MCP-grounded live-registry check exists alongside it (see "Where this plugs into the pipeline").

Demo: it actually runs

```
$ scripts/search_pipeline_catalog.py \  
  "Illumina paired-end amplicon sequencing of SARS-CoV-2, \  
  want variant calls and lineage assignment with pangolin"  
  
1. [10 matched terms] SARS-CoV-2: Illumina Amplicon PE (iVar/pangolin/nextclade)  
   matched on: amplicon, assignment, cov, end, illumina,  
               lineage, paired, pangolin, sars, variant  
  
2. [8 matched terms] SARS-CoV-2: WGS PE Variant Calling  
3. [7 matched terms] SARS-CoV-2: Illumina Amplicon PE (lofreq, ARTIC)
```

Correctly ranks the pangolin/nextclade workflow first — it's the only candidate matching "assignment," "lineage," **and** "pangolin."

Demo: it also knows when it doesn't know

```
$ scripts/search_pipeline_catalog.py \  
  "differential expression analysis of mouse RNA-seq data"
```

```
No catalog matches. This is a candidate for the 'build' path,  
or a sign the catalog needs a new entry -- not a silent failure.
```

This matters for **Goal 5 (adversarial review)**: a router that always returns *something* invites confident fabrication. Returning nothing when nothing matches is the correct, honest behavior.

Known limits of the v0 matcher

- **Synonym-blind:** "flu" only matches because it's hand-listed as a keyword alongside "influenza" — a paraphrase not anticipated will miss
- **No organism-name specificity:** a request naming an organism not in any keyword list (most named bacteria/pathogens) won't match even where a genus-agnostic workflow (e.g. AMR detection) would actually apply
- **This catalog itself is a curated ~20-workflow subset** — but no longer the whole story: `route.py` now also checks the **full, live IWC registry** via a real MCP client (`scripts/galaxy_mcp_client.py` → [galaxy-mcp](#)) whenever this catalog finds nothing. That gap is closed — see the next slide.

Still genuinely open: the matching itself, at both layers, is keyword-based (IDF-weighted overlap here, BM25 live) — synonym/paraphrase-blindness remains until embedding-based semantic search replaces either one.

Where this plugs into the pipeline

```
interview (free-text) → search_pipeline_catalog.py → route.py
                        |
                        ranked candidates + matched terms
                        |
                        use / adapt / build decision (Goal 1)
                        |
                        (if "build") check the live IWC
                        registry via galaxy_mcp_client.py
                        → galaxy-mcp – surfaced for review,
                        never auto-upgrades the decision
```

- **Confident top match** → propose **use** or **adapt**
- **No match, or all low-confidence** → check the live registry first;
still nothing → **build**, and log it for the Goal 7 pass-rate/
failure taxonomy

Files: `knowledge_base/pathogen_genomics/galaxy_pipeline_catalog.yaml` ·
`scripts/search_pipeline_catalog.py` · `scripts/galaxy_mcp_client.py`